

MATH 189 - Data Analysis

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1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.

"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, \dots, X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$

We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of } X \end{cases}$

Question: Why estimate f ?

Prediction: Given X , we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned} \mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x))^2 \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2) \end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = (f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

”Inference” - Sometimes we want to understand the association between y and $X_1, \dots, X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where $X_i = (X_{i1}, X_{i2}, \dots, X_{ip})$

$n = \#$ data points (e.g. $n = 30$)
 $p = \#$ predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

”Fit/train the model” usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, \dots, \beta_p$ that make the MSE and hopefully the test MSE small.

Classification

Data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where $X_i = (x_{i1}, \dots, x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)

The KNN classifier is defined by letting $\eta_0 =$ set of indices $i \in \{1, \dots, n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j